

Solution:

- (a) Define direction vector $s = (s_1, \dots, s_d) \in \{+1, -1\}^d$, and projection

$$f_s(p) = \sum_{i=1}^d p_i s_i$$

An observation is that for any direction vector we have:

$$\sum_{i=1}^d |p_i - q_i| = \max_{s \in \{-1, 1\}^d} \sum_{i=1}^d s_i (p_i - q_i)$$

because for each i one can choose s_i to be $+1$ or -1 so that $s_i(p_i - q_i) = |p_i - q_i|$. Thus,

$$D(p, q) = \max_{s \in \{-1, 1\}^d} (f_s(p) - f_s(q)).$$

For each of the 2^d choices of $s \in \{+1, -1\}^d$ we scan once through all n points and find candidate points that could be the farthest from q :

$$M_s = \max_{p \in P} f_s(p), \quad m_s = \min_{p \in P} f_s(p).$$

Once we know M_s and m_s , the best choice of $f_s(q)$ is to place

$$f_s(q) = \frac{M_s + m_s}{2}.$$

Each such scan takes $O(nd)$ time (to form each dot product $f_s(p)$ in $O(d)$ and take a max/min), so over all s this is

$$O(2^d \cdot n \cdot d).$$

Therefore, the running time grows exponentially with d . For any constant dimension d , the problem can be solved in $O(n)$ time.

(b) **Construct LP:**

For each input point $p^{(j)} = (p_1^{(j)}, \dots, p_d^{(j)})$, and for each coordinate $i \in \{1, \dots, d\}$, introduce a variable $t_{j,i} \geq 0$ that will represent $|q_i - p_i^{(j)}|$.

Introduce a single variable $r \geq 0$ to represent the maximum rectilinear distance $\max_j D(p^{(j)}, q)$.

In total, there are $O(nd)$ variables.

For each point $j \in \{1, \dots, n\}$ and each coordinate $i \in \{1, \dots, d\}$, enforce

$$\begin{cases} q_i - p_i^{(j)} \leq t_{j,i}, \\ p_i^{(j)} - q_i \leq t_{j,i}. \end{cases}$$

These two inequalities together force $t_{j,i} \geq |q_i - p_i^{(j)}|$.

For each point j , its rectilinear distance to q is $\sum_{i=1}^d |q_i - p_i^{(j)}|$. We enforce

$$\sum_{i=1}^d t_{j,i} \leq r.$$

This guarantees that r is at least as large as every $D(p^{(j)}, q)$.

That is $2nd$ constraints from the absolute-value linearization, plus n constraints from the sum $\leq r$ requirements, for a total of $O(nd)$ constraints.

The objective of this LP is to minimize r .

Proof of correctness:

($\implies$) center with radius r :

Any choice of q and r that satisfies the LP constraints (i.e., the rectilinear 1-center) must have $\sum_i t_{j,i} \leq r$ for each j , and each $t_{j,i} \geq |q_i - p_i^{(j)}|$. Hence $\sum_i |q_i - p_i^{(j)}| \leq \sum_i t_{j,i} \leq r$, i.e. $D(p^{(j)}, q) \leq r$ for all j . Conversely, any q with $\max_j D(p^{(j)}, q) \leq r$ gives a feasible LP solution by setting $t_{j,i} = |q_i - p_i^{(j)}|$.

($\impliedby$) minimal max-distance:

Since the LP minimizes r , the optimum LP value r^* is the smallest radius for which a center q exists that keeps all points within distance r^* . That is exactly the rectilinear 1-center objective.

Time complexity:

There are $O(nd)$ variables and $O(nd)$ constraints, hence we can solve the LP problem in $\text{poly}(n, d)$ time.